Lab Assignment 3: Operational Amplifier Applications

Revision: September 20, 2018

Summary

In this lab, we use operational amplifiers (op-amps) to design, analyze and implement some simple but practical circuits. These circuits include:

- A first-order active high pass filter (in contrast with the passive low pass filters investigated in lab assignment 1);
- An op-amp amplifier with high gain, high input resistance, and low output resistance.

Learning Outcomes: After completing this lab, you should be able to:

- Design active filters to perform high pass filtering operations.
- Design an op-amp amplifier with specified gain, input resistance, and output resistance.

Required Equipment

- EE352 analog parts kit
- Breadboard
- Function generator, oscilloscope
- DMM
- DC power supply

I. First Order High Pass Active Filter

A common use of op amps is to implement filtering operations. In Lab 1, we constructed a filter out of passive components to perform low pass filtering. In this portion of the lab assignment, we will design a system to implement high pass filtering using active components.

Pre-lab:

(a) Derive the differential equation for the output voltage of the circuit shown in Figure 1. Assume that the op-amp is ideal. Also find the initial conditions, assuming zero initial charge on the capacitor at time 0–.
(b) Using your results from part (a), find an analytical expression for the step response of the circuit. What is the time constant τ for this circuit?
(c) Find an analytical expression for the amplitude response of the circuit, assuming an ideal op amp. Check your solution for very low and very high frequency inputs; do the responses predicted by your equations match your physical understanding of the circuit?
(d) Determine the circuit's cutoff frequency ω_c. What is the relationship between ω_c and τ?
(e) Design a high pass filter based on the circuit of Figure 1 with an input resistance of 10 KΩ, a high frequency gain of 6 dB, and a –3 dB (cutoff) frequency of approximately 10 KHz. Choose realistic resistor and capacitor values; you will be implementing your design in the lab.
(f) Use PSPICE (LTSPICE or Orcad) to simulate the frequency response and step response of your circuit. If you use Orcad, then use the OP-27 from the OPAMP library.

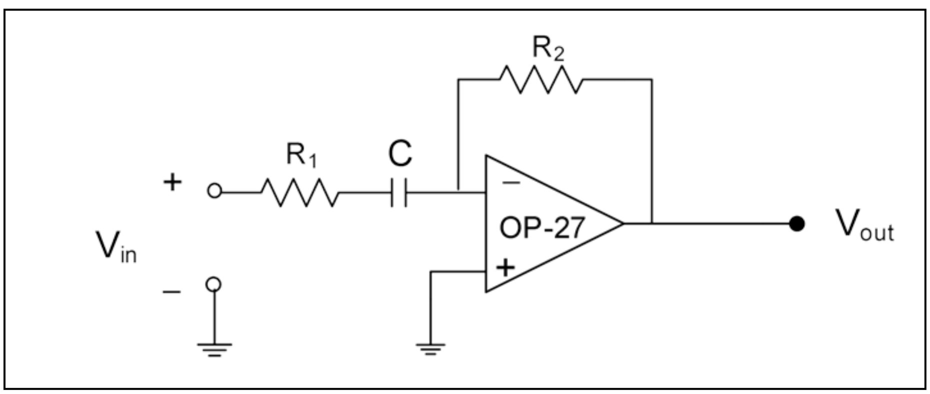


Figure 1. First order op-amp high pass filter.

Lab Procedures:

1. Construct the circuit you designed in the pre-lab.
2. The OP-27 V+ and V– power inputs should be connected to the power supply +12 and –12 volt outputs. *Before powering up your circuit, please have the TA verify that you have set up the power supply correctly to supply ±12 volts – an incorrectly configured power supply can result in a burned out op amp.*
3. *Step Response:* Using the signal generator input a 1V p-p square wave to your circuit; adjust the offset so that the wave varies between 0 volts and 1 volts. Hook the oscilloscope to the output of your filter to measure the step response. Adjust the square wave frequency to see the entire step response curve. Save your step response on a USB drive. Using the

cursors, compute the time constant τ for your circuit. Does your step response match your PSPICE simulation? Is the value of τ as expected?

4. *Frequency Response:* Using a 4V p-p sine wave from the signal generator, measure the circuit's amplitude frequency response at a minimum of 10 different frequencies, spanning a frequency range of at least an order of magnitude below the cutoff frequency to an order of magnitude above the cutoff frequency. You must obtain a smooth curve similar to the simulated results, and explain the method used to locate the cutoff frequency. Does your measured frequency response match the PSPICE simulation? Do the high frequency gain and the cutoff frequency both conform to the design specifications?
5. *Demonstrate your circuit's step response and frequency response to the TA and have him/her initial your lab checklist. Compare your measured responses with your PSPICE results.*

II. <u>High Gain Amplifier</u>

In this section of the lab assignment, we will design and implement a high-gain amplifier with high input resistance and low output resistance.

Design specifications:

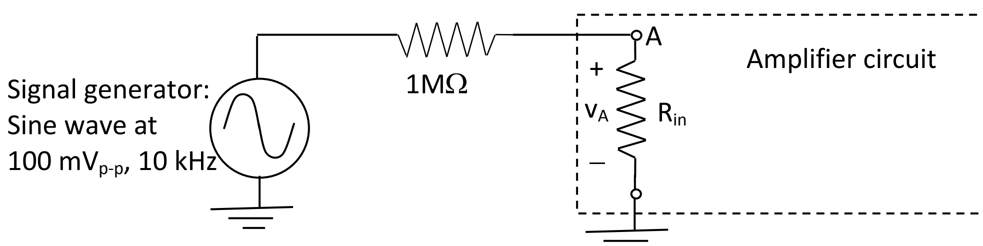
1. Input signal: 10 kHz sine wave; adjust the amplitude to be between 6 to 20 mV peak-to-peak.
2. Amplifier gain: 1000±2%.
3. Output signal: 10 kHz sine wave, amplitude 1000 times input amplitude, with minimum phase shift with respect to input. In particular, a 180 degree phase shift between input and output is not acceptable.
4. Measured input resistance: greater than or equal to 1 MΩ.
5. Measured output resistance: less than or equal to 10 Ω.
6. Op-amp: Use the same OP-27 op-amps used in Part I of this lab.
7. Use not more than one DC power supplies (Rigol DC power supply) to implement your design. Power to the op-amp's V+ and V– inputs should not exceed positive and negative 12 volts. (These power supplies provide two variable power supplies and one fixed power supply – this should be more than adequate for this problem.)

Circuit design procedure:

1. Read Module 3.3 (available on the course website) before beginning the circuit design.
2. Typical implementations of this amplifier consist of between two and five op-amp stages. The design approach is, however, up to you – design the circuit any way you wish.
3. For your designed circuit, calculate theoretical values for the expected gain, and the expected input and output resistances, given the input and output signals described above.
4. If your circuit has multiple stages, you should also calculate the expected gain, and expected input and output resistances for each stage. *When designing multi-stage amplifiers, it is very important that the input resistance of the next stage be much higher than the output resistance of the current stage.*
5. It is also important to note that at the input frequency of 10 kHz, the open-loop voltage gain of your OP-27 op-amp is about 1000. This means that you will not be able to use only one op amp for your amplifier design. The open-loop gain decreases with frequency, as shown in OP-27 spec sheet available on the course website.
6. You are encouraged to use PSPICE to test and verify your design, but that is not required.

Lab Procedures:

1. Implement your designed high gain amplifier circuit. If your circuit has multiple stages, it is recommended that you implement and test the first stage, then implement and test the second stage, then connect the first and second stages and test them together, etc.
2. To eliminate DC offsets, you may find it useful to put a coupling capacitor between the output of each amplifier stage and the input of the next. This is particularly important for the final stages of your amplifier, where the input might have a significant DC offset which could prevent your amplifier from reaching its full gain. Because the series combination of your coupling capacitor C and the resistance R_{in} of the next stage gives a high pass filter with cutoff frequency $\omega_c = 1/(R_{in}C)$, you should choose R_{in} and C for each stage so that ω_c is higher than your signal frequency of 10000 Hz. *Note, however, that the very low DC offset of the OP27 op-amp may allow your circuit to work fine without any coupling capacitors.*
3. To reduce noise in your amplifier, it often helps to place a capacitor (the value should be between 1 and 10 uF) between V+ and ground, and another such capacitor between V– and ground. If you run V+ and V– on the common power rails on your breadboard, then only two capacitors (one for V+, and one for V–) will be needed. *If you use electrolytic capacitors (the black can-shaped capacitors in your kit), then make sure the polarity is hooked up correctly; for example, the – terminal of the capacitor should hook to V–, and the + terminal to ground. Hooking up electrolytic capacitors with the wrong polarity can result in an explosion and possible injuries.*
4. Measure and record in your lab notebook the overall gain of the circuit for the input conditions given in the design specifications. It may be necessary to use a voltage divider to produce the 6 to 20 mV p-p input signal, if your signal generator cannot produce such a small-amplitude signal. To measure the input signal, you should use a 1x scope probe, and put the vertical scale of the scope on 2 mV per division. Use the Acquire key to set signal averaging to at least 16 to reduce noise on the small input signal.
5. *Measure and record the input resistance:* To measure the input resistance, hook up the circuit shown below. The point labeled A is the input to your amplifier, and the resistor labeled R_{in} is the equivalent input resistance of your amplifier – not an actual resistor.



   Measure the voltage v_A at your circuit's input terminal A with the scope, using the 10x scope probe (which has the highest input impedance), and find the value of R_{in} from v_A; record the value of R_{in} in your lab notebook. If your scope probe has infinite input resistance, then you would expect v_A be ≥ 50 mV p-p if R_{in} is ≥ 1 MΩ. You may want to set up a test voltage divider with two 1 MΩ resistors to verify that your 10x scope probe gives you the voltage measurement that you expect. (Note: At 100mV p-p input, your amplifier's output will saturate at +– 12 V; that will not affect your measurement of R_{in}.)
6. *Measure and record the output resistance:* First measure V_{oc}, the open circuit voltage at the output of your amplifier, when the input is a 6 to 10 mV p-p 10 kHz sine wave. Then measure V_L, the output voltage when a 470Ω load resistor is connected between your amplifier output and ground. If $V_L > (470/480) \times V_{oc}$, then your circuit's output resistance R_{out}

is < 10 Ω. From your measurements of V_{oc} and V_L, find the value of R_{out} and record it in your lab notebook. (Note: If R_{out} is larger than it should be and the amplifier input is at 10 mV p-p, try turning the input down to 6 mV p-p, and then re-measure R_{out}. The OP27 has an absolute output current limit of 30 mA typical, but some op-amps might current limit at less than the typical value.)

7. Verify that the measured gain, input and output resistances meet the system requirements, and that they are consistent with the expected values you calculated when you designed the circuit.
8. *Demonstrate operation of the circuit to a teaching assistant and have him/her initial your lab checklist.*

Lab Report:

In your lab report, provide a summary of the results of this lab assignment. You should include, at a minimum, all items indicated on the lab checklist. Append the lab checklist sheet with teaching assistant initials indicating completed lab demos to your report.

Lab 3 Checklist

Name:__________________

Lab title and introduction (5 pts)

- Lab title, your name, date, and lab partner.
- Brief introduction (one to three sentences) explaining the purpose of this lab.

I. First Order High Pass Active Filter (45 pts total)

1. Differential equation relating v_{out} and v_{in} for circuit of Figure 1. (3 pts)
2. Selected values for R_1, R_2, and C to meet design specifications. (3 pts)
3. PSPICE simulation of circuit's frequency response. (3 pts)
4. PSPICE simulation of circuit's step response. (3pts)
5. Measured values for R_1, R_2, and C used in circuit implementation. (3 pts)
6. **DEMO**: Have a teaching assistant initial this sheet, indicating that they have observed your circuit's operation. (20 pts)
    - Circuit sinusoidal operation is observed for at least one value of input frequency
    - Circuit design parameters R_1, R_2, and C are appropriate (show TA analysis from pre-lab providing design approach).
    - Discuss with TA whether circuit's frequency response is consistent with expectations from pre-lab analysis and PSPICE simulations.
    - Discuss with TA whether circuit's step response is consistent with expectations from pre-lab analysis and PSPICE simulations.
7. Plot your experimental frequency response data in your lab report. Also provide either a sketch or a picture of your step response. Compare your experimental data and simulation results and discuss any discrepancies. Does your circuit's response meet the design specifications? (10 pts)

II. High Gain Amplifier (50 pts total)

(a) Circuit Design (20 pts total)

1. Sketch of amplifier circuit. Include desired values for circuit components (e.g. resistor values). (10 pts)
2. Expected gain, input and output resistances for your overall amplifier circuit, and for each stage within your amplifier circuit. (Hand calculations using information in Module 3.3). (10 pts)

(b) Circuit Implementation and testing (30 pts total)

1. Measured values of circuit components used in your design. (2 pts)
2. Measured values of the circuit's gain, input resistance, and output resistance. Does the circuit meet specifications? (8 pts)
3. **DEMO**: Have a TA initial this sheet, indicating that he/she has observed (i) your circuits' operation; (ii) your measurements verifying that the circuit meets the design specifications, and (iii) your calculations of gain and input and output resistance for the overall circuit and for each stage within the circuit. (15 pts)
4. Comparison of measured and expected gain, input and output resistances for your circuit and discussion of any discrepancies between measurements and expectations. (5 pts)

